FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10 — SUB-STRAND 1.1: INTRODUCTION TO CHEMISTRY	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You have spent 10 lessons exploring Chemistry through the lens of one familiar, everyday event — chapati dough transforming on a jiko. Now it is time to pull everything together and show what you have learned.

Your goal is to write a detailed, well-organised explanation that answers the driving question:

"How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?"

READ THIS CAREFULLY BEFORE YOU BEGIN:

– Answer every section prompt fully. Use scientific vocabulary you have learned in this unit.
– Connect your ideas back to the chapati phenomenon wherever you can — it is your anchor.
– Use real Kenyan examples (food, farming, industry, health) to support your points.
– Write in full sentences and organise your ideas clearly.
– Your explanation should read as one connected argument, not a list of facts.
– Aim for depth over length — a focused, evidence-based explanation is better than a long, vague one.

Section 1: Defining Chemistry Through the Chapati Phenomenon	
Define Chemistry in your own words and explain which branch (or branches) of Chemistry best explains what happens when chapati dough cooks on a jiko. In your answer: (a) state what Chemistry is and what it studies, (b) name at least THREE branches of	Chemistry is the branch of science that studies matter — what it is made of, how it is structured, and how it changes. When we watched chapati dough being placed on a hot jiko and transform into a firm, golden disc, we were witnessing Chemistry in action. Chemistry has several branches, each focusing on a different aspect of matter and its changes. Organic Chemistry studies compounds that contain carbon, including the carbohydrates and proteins found in wheat flour. Inorganic Chemistry deals with non-carbon-based substances, such as the mineral salts in our food. Physical Chemistry examines the relationship between energy and matter — it explains how heat from the jiko drives chemical reactions. Biochemistry explores chemical processes inside living organisms, such as how our bodies digest the chapati we eat. Analytical Chemistry is concerned with identifying the composition of

Chemistry and briefly describe each one, and (c) explain which branch(es) directly apply to the chapati transformation and why. Use scientific vocabulary such as 'matter', 'composition', 'properties', 'change', and 'energy'.	substances, which is how food scientists test whether a chapati is safe to eat. The branches most directly relevant to the chapati transformation are Physical Chemistry and Organic Chemistry. Physical Chemistry explains how thermal energy from the jiko raises the temperature of the dough, providing the activation energy needed to break and form chemical bonds. Organic Chemistry explains the changes happening inside the dough itself — the proteins in the wheat flour (gluten) denature and set, while the starch granules gelatinise and then firm up. The Maillard reaction, a chemical reaction between amino acids and reducing sugars, produces the golden-brown colour and the appealing aroma we associate with freshly cooked chapati. These are not physical changes — they are chemical changes, meaning the composition and properties of the matter have fundamentally changed.

Section 2: Physical vs. Chemical Change — Why You Cannot Un-cook a Chapati	
Explain the difference between a physical change and a chemical change, and use this to explain why the chapati transformation is irreversible. In your answer: (a) clearly distinguish between physical and chemical changes, giving one everyday Kenyan example of each (other than chapati), (b) identify at least THREE pieces of evidence from the chapati phenomenon that confirm a chemical change has taken place, and (c) explain what 'irreversible' means at the molecular level — what has actually changed in the matter that makes it impossible to get the dough back?	A physical change alters the appearance or state of a substance but does not change its chemical composition. For example, when you dissolve sugar in chai (tea), the sugar disappears into the liquid, but it is still sugar — you could, in theory, evaporate the water and recover the sugar crystals. No new substance is formed. A chemical change, by contrast, produces one or more new substances with different compositions and properties from the original. When maize is fermented to make uji or busaa, new compounds — including organic acids and alcohols — are produced. This is a chemical change. The chapati transformation shows at least three clear signs of a chemical change. First, there is a change in colour: the pale, off-white dough becomes a golden-brown disc — evidence of new pigment compounds being formed in the Maillard reaction. Second, there is a change in texture and physical properties: the soft, elastic dough becomes firm and rigid because the gluten proteins have denatured and the starch has gelatinised — these are new structural arrangements of molecules. Third, a new aroma is produced: the appealing smell of cooked chapati comes from volatile organic compounds that were not present in the raw dough. At the molecular level, 'irreversible' means that the chemical bonds within the molecules have broken and new bonds have formed, creating entirely new substances. The long protein chains in the dough have unfolded and cross-linked with each other in new configurations. The starch granules have burst and re-set. You cannot simply remove heat and expect the molecules to reassemble into their original form — the energy required to reverse these specific bond changes under everyday conditions is not available. This is why, no matter how long you wait, the chapati will never become dough again.

Section 3: The Importance of Chemistry in Kenyan Industry and Daily Life	
Explain how Chemistry — and an understanding of chemical changes — is important in at least THREE areas of daily life or industry in Kenya. For each area, describe: (a) what chemical process or principle is involved, and (b) why understanding that chemistry	An understanding of Chemistry is essential in many areas of Kenyan life and industry. First, in food science and production, Chemistry explains every transformation from raw ingredient to finished product. The same chemical principles that explain the cooking of chapati — protein denaturation, starch gelatinisation, the Maillard reaction — apply to the processing of maize into ugali, the fermentation of milk into mala (sour milk), and the preservation of fish from Lake Victoria. When food technologists at the Kenya Bureau of Standards (KEBS) use Analytical Chemistry to test food products, they are checking that the chemical composition meets safety standards. Without this knowledge, harmful substances could enter the food supply.

leads to better, safer, or more efficient outcomes. You may draw on examples from food production, agriculture, medicine, manufacturing, or environmental protection. Connect at least one example back to the chapati phenomenon or the broader theme of matter and its changes.	Second, in agriculture, Chemistry underpins soil science, fertiliser production, and crop protection. Farmers in the Rift Valley and Kericho use nitrogen-based fertilisers whose effectiveness depends on chemical reactions in the soil. Understanding chemistry helps farmers choose the right fertiliser for their soil type, avoid over-application that causes chemical runoff into rivers and lakes, and use pesticides safely. Inorganic and Physical Chemistry explain how these compounds interact with soil particles and water. Third, in medicine and public health, Chemistry is the foundation of pharmaceutical science. Every drug — from paracetamol for fever to antimalarials like artemether — is a chemical compound whose effect on the body depends on its molecular structure and how it reacts with biological molecules. Kenyan health professionals and patients who understand basic Chemistry are better placed to use medicines correctly, recognise harmful drug interactions, and avoid counterfeit medicines. This directly connects to our driving question: understanding Chemistry is not just academic — it enables us to make safer, smarter choices every day.

Section 4: Chemistry, Drug Use, and the Human Body	
Explain how Chemistry helps us understand the effects of drugs and substances on the human body, and why this knowledge is important for making healthy choices. In your answer: (a) define what a drug is from a Chemistry perspective, (b) explain — using chemical reasoning — how a substance introduced into the body can change the way the body's own chemistry works (you may refer to concepts such as chemical reactions, molecular interactions, or changes to normal body processes), and (c) use at least ONE specific example of a harmful substance (such as alcohol, tobacco, or an illicit drug) and explain the chemical harm it causes. Connect your answer to the broader theme of irreversible chemical changes.	From a Chemistry perspective, a drug is any chemical substance that, when introduced into the body, alters normal biological chemical processes. Just as heat causes irreversible chemical changes in chapati dough, some drugs can cause irreversible or difficult-to-reverse changes in the body's own chemical systems. The human body runs on chemistry — enzymes catalyse reactions, hormones carry chemical signals, and neurons communicate through neurotransmitters that are released and received through molecular interactions. When a foreign chemical substance enters this system, it can interfere with these processes. Some drugs mimic the shape of natural molecules, binding to receptors and triggering responses the body did not intend. Others block receptors, preventing normal signals from getting through. In both cases, the body's carefully balanced chemistry is disrupted. Consider alcohol (ethanol, C_2H_5OH). When consumed, ethanol is absorbed into the bloodstream and crosses the blood-brain barrier. In the brain, it enhances the effect of the inhibitory neurotransmitter GABA and suppresses the excitatory neurotransmitter glutamate. This slows brain activity, impairing judgment, coordination, and reaction time. In the liver, enzymes break ethanol down into acetaldehyde — a toxic compound that damages liver cells. Long-term alcohol use causes liver cells to die and be replaced by scar tissue (cirrhosis), a largely irreversible chemical and structural change. Just as we cannot un-cook a chapati, we cannot un-scar a liver. Understanding these chemical mechanisms is not meant to frighten us — it is meant to empower us. A student who understands that alcohol causes real, measurable, sometimes irreversible chemical damage to the brain and liver is better equipped to make an informed choice than one who has only been told 'it is bad for you.' Chemistry gives us the language and the evidence to make truly informed decisions.

Section 5: Answering the Driving Question — Your Final Argument

Now bring everything together. Write a concise but complete answer to the driving question: 'How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?' Your answer should: (a) summarise how Chemistry — through its definition, branches, and key concepts — explains everyday changes in matter (use the chapati as your final connecting example), (b) explain the link between scientific understanding and real-world decision-making, and (c) reflect on what has changed in YOUR own thinking about Chemistry since Lesson 1. What did you think Chemistry was before? What do you think now?	Chemistry is the science of matter and its changes — and matter is everything around us. From the chapati cooking on a jiko in a Nairobi kitchen, to the fertiliser applied to a maize farm in the Rift Valley, to the painkiller dissolving in a glass of water, Chemistry is at work in every moment of our lives. The different branches of Chemistry — Organic, Inorganic, Physical, Biochemistry, and Analytical — each shine a different light on these changes, helping us understand not just what happens, but why and how. The chapati phenomenon taught us something profound: some changes in matter are irreversible. The dough cannot become dough again because new chemical bonds have formed and new substances have been created. This same principle applies to the human body. Chemical substances — whether the food we eat, the medicines we take, or the drugs we might be pressured to try — all interact with our body's chemistry. Some of those interactions, if we are not careful, can cause changes that are very difficult or impossible to reverse. Understanding Chemistry therefore does not just help us pass an exam. It helps us ask better questions: Is this substance safe? What will it do inside my body? Has this food been processed in a way that changes its nutritional value? Why does this medicine have side effects? These are Chemistry questions — and people who can answer them are safer, healthier, and more empowered than those who cannot. At the start of Lesson 1, many of us thought Chemistry was about laboratories, equations, and complicated formulas that had little to do with our daily lives. Now we understand that every meal we cook, every medicine we take, every crop our families grow, and every choice we make about substances we put into our bodies is a Chemistry decision. The jiko in a Kenyan kitchen is a chemistry laboratory. We are all, already, practising chemists — and the more we understand the science behind what we observe, the better choices we can make for ourselves, our families, and our communities.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Understanding of Chemistry: Definition, Branches, and Matter	Accurately defines Chemistry and clearly explains at least three branches with correct descriptions. Correctly distinguishes between physical and chemical changes using precise scientific vocabulary (matter, composition, properties, bonds, energy). All explanations are directly linked to the chapati phenomenon and additional Kenyan examples.	Defines Chemistry correctly and names at least two branches with mostly accurate descriptions. Distinguishes between physical and chemical changes with minor gaps. Uses some scientific vocabulary. Most explanations are connected to the chapati phenomenon.	Provides a vague or partially correct definition of Chemistry. Names branches but descriptions are inaccurate or confused. Struggles to clearly distinguish between physical and chemical changes. Little scientific vocabulary used. Weak or absent connection to the chapati phenomenon.
Explanation of the Chapati Phenomenon as a Chemical Change	Identifies at least three distinct pieces of evidence of chemical change in the chapati phenomenon (e.g., colour change, new aroma, irreversible texture change). Explains irreversibility at the molecular level (bond breaking/forming,	Identifies two pieces of evidence of chemical change and provides a reasonable explanation of irreversibility. Molecular-level explanation is present but may lack full detail. Explanation is mostly accurate.	Identifies only one piece of evidence or confuses physical and chemical change. Explanation of irreversibility is vague or relies only on everyday language without scientific reasoning. Little or no molecular-level thinking shown.

	new substances formed). Explanation is scientifically accurate and clearly articulated.		
Application of Chemistry to Kenyan Daily Life and Industry	Clearly explains Chemistry's importance in at least three distinct areas of Kenyan life or industry (e.g., food, agriculture, medicine). For each area, correctly identifies the relevant chemical process and explains why chemical understanding leads to better outcomes. Examples are specific, Kenya-relevant, and well-integrated into the argument.	Explains Chemistry's importance in two or three areas with mostly correct chemical reasoning. Examples are relevant but may be generic or lack depth. The link between chemistry knowledge and improved outcomes is stated but not fully developed.	Mentions one or two areas but explanations are vague or incorrect. Little chemical reasoning is applied. Examples are weak, irrelevant, or absent. The link between chemistry and better outcomes is not clearly made.
Chemistry of Drug Use and Health Decision-Making	Accurately defines a drug from a Chemistry perspective. Clearly explains, using chemical reasoning, how substances alter the body's normal chemical processes. Provides a specific, scientifically accurate example of a harmful substance and explains the chemical damage it causes. Explicitly connects drug harm to the concept of irreversible chemical change. Demonstrates mature, evidence-based reasoning about health choices.	Defines a drug adequately and gives a reasonable chemical explanation of how it affects the body. Example of harmful substance is mostly accurate. The connection to irreversible change is present but may be underdeveloped. Reasoning about health choices is evident.	Definition of a drug is vague or purely biological without chemical reasoning. Explanation of how substances affect the body lacks chemical thinking. Example of harmful substance is vague or the chemical harm is not explained. Little connection to irreversible change or informed decision-making.
Quality of Final Argument and Reflection	The final section provides a compelling, well-structured answer to the driving question that integrates all major concepts from the unit. The chapati phenomenon is used as a coherent connecting thread throughout. The student reflects meaningfully and specifically on how their thinking has changed, demonstrating genuine conceptual growth. Writing is clear, confident, and uses accurate scientific language throughout.	The final section answers the driving question with most major concepts included. The chapati connection is present. Reflection on personal learning is genuine but may be brief or general. Writing is mostly clear with adequate use of scientific language.	The final section partially addresses the driving question but key concepts are missing or superficially treated. The chapati connection is weak or absent. Reflection is minimal or does not demonstrate clear conceptual growth. Writing is unclear or scientific language is rarely used.